PD-TAZ-WH-1

PRINTDECK

UPGRADE KIT

For the LulzBot TAZ Workhorse

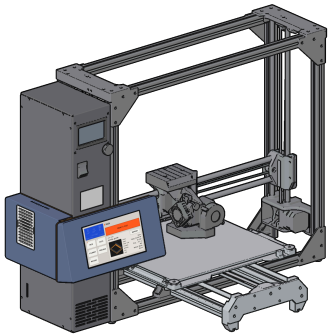


Printdeck is free software: AGPL-3.0, with the Go port of Klipper under GPL-3.0.
The deck draws every picture of this booklet.

https://github.com/beetlebugorg/printdeck

INTRODUCING PRINTDECK FOR THE TAZ WORKHORSE

Your Workhorse called. It wants a dashboard.



NOW WITH TOUCH!

Let's be honest. Your TAZ Workhorse works fine. It prints. It has a knob. Nobody asked you to change a thing.

That is exactly why you should.

Printdeck puts a seven-inch touch screen on the front of the printer, in a case that you print on the very printer you upgrade. The printer builds its own upgrade. If that is not the future, we do not know what is.

Inside is Klipper, ported to Go, in one program. It plans every move with the same look-ahead as the printer, so the time left is the real time left. Not a guess. Not a mood.

It updates itself. It flashes the firmware of the printer for you. It even takes over the old display of the printer and puts the same menus on it, so the knob gets a promotion too.

No drilling. No nuts in the rail. Two cover screws and two printed clips. If you can turn a hex key, you are qualified.

- ✔ 7-inch color touch screen, mounted wide
- ✔ The Klipper host, in Go, in one program
- ✔ Time left from the motion planner
- ✔ Signed updates that install themselves
- ✔ 344 g of PLA, 16 h 42 min, no supports

✂

YES! Upgrade my Workhorse. I have no good reason.

□ Because I can □ Because I can □ Both

Name _____ Printer _____

Mail to: your own printer. Allow 16 h 42 min for delivery.

Printdeck is free software (AGPL-3.0). Printer not included. LulzBot and TAZ Workhorse are names of their owner. Printdeck is not made by LulzBot and is not connected with LulzBot. The time left is exact for the moves of the file; the heating and your pauses are your own business.

Notes

Settings

The file printdeck.txt on the PRINTDECK partition holds the settings. Put the disk in a computer to change it. The deck reads it at each start, then clears the Wi-Fi password in it.

`wifi_ssid, wifi_password, wifi_country`	The Wi-Fi network
`hostname, timezone`	The name on the network, and the time zone
`display_rotate`	90 or 270 for a display that is mounted tall, 0 or 180 for one that is mounted wide
`touch_invert_x, touch_invert_y, touch_swap_xy`	1 turns an axis of the touch screen
`preheat_tool, preheat_bed`	The temperatures of Preheat with no filament set
`night_from, night_to`	The hours of the night colors
`printer.cfg (a file)`	A Klipper config. The deck takes it at the next start.

Contents

UPGRADE THE WORKHORSE
What you build 4
Print the parts 5
What you need 7
Before you start 9
Prepare the drive 11

BUILD THE MOUNT
Build the mount 12
Connect and start 24

SET UP
Connect to Wi-Fi 26
Choose your printer 28
Install the Klipper firmware 30

MEET YOUR PRINTDECK
The deck 31
The display of the printer 33
The deck in a browser 34

PRINT A PART
Send a file from the slicer 35
Start a print 36
Watch the print 38
Print in two colors 45
Look around during a print 47

YOUR IPHONE AND APPLE WATCH
The iPhone and Apple Watch apps 48

CHANGE THE FILAMENT
Change the filament 50

MOVE AND HEAT
Move the head and heat 56

LEVEL AND CALIBRATE
Level and calibrate 58

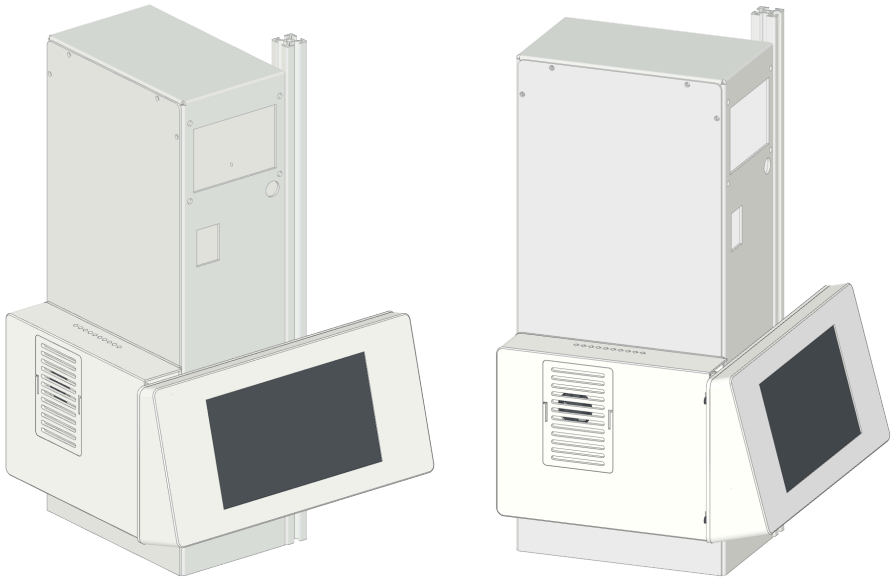
LOOK AFTER THE PRINTER
Look after the printer 63
Update the Printdeck 68

WHEN SOMETHING GOES WRONG
When something goes wrong 69
Settings 70

What you build

This kit puts the Printdeck on the front left of your TAZ Workhorse. The 7 inch touch screen is in a wide head below the TAZ Workhorse logo, and the Raspberry Pi 5 is in a pod on the left side of the electronics chassis. The head leans back 15 degrees, so you see the screen from the front of the printer.

You print the parts on the Workhorse itself. Two printed clips hold on the front rail with metal T-nuts, and two screws of the side cover hold the plate. You drill no hole in the printer.



`8 printed parts` `344 g of PLA` `16 h 42 min of printing`
`No supports` `No holes in the printer`

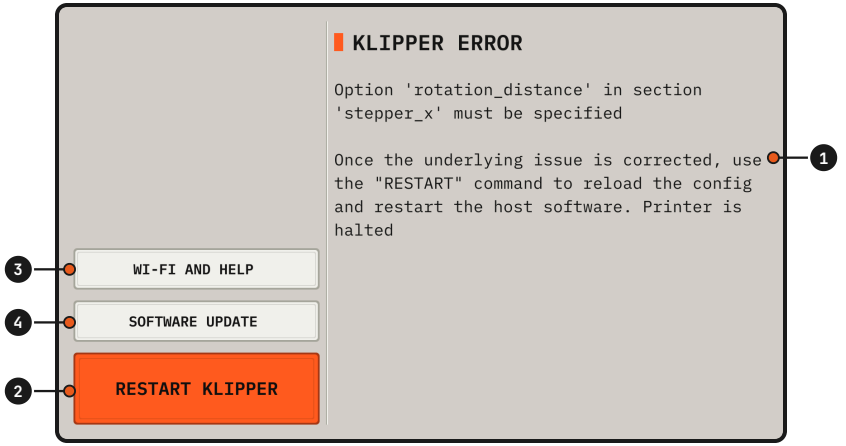
The deck screens in this booklet are the wide screens: the display is mounted wide in the head.

When something goes wrong

When Klipper stops with an error, the deck shows all the lines of its message.

1. Read the message. It often names the setting in printer.cfg.
2. After a stop or a power cut of the printer, tap Firmware restart.
3. After a change to printer.cfg, tap Restart Klipper.

A Klipper error



1. **Message.** What Klipper reports, with all its lines.
2. **Restart Klipper.** Restarts Klipper. It reads printer.cfg again.
3. **Wi-Fi and help.** The network, the address and the password of the deck.
4. **Software update.** Installs a new version of Printdeck OS. A new version can fix the problem.

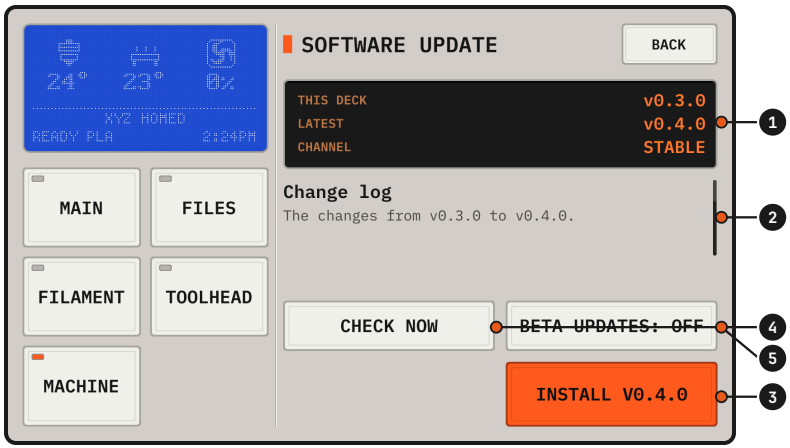
Update the Printdeck

The deck checks for a new version once a day. Machine then shows Software update in orange.

Each version has a signature. The deck installs only a version with a valid signature of Printdeck. It writes the other system slot and restarts into it. If the new system does not start, the deck goes back to the old one.

1. Open Machine, then Software update.
2. Read the notes of the new version, then tap Install. The deck does not install during a print.
3. Wait while the deck downloads and writes the update. It restarts into the new version.

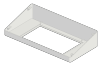
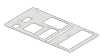
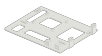
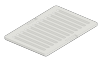
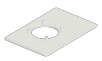
Software update



1. **Versions.** The version of this deck, the newest version, and the channel.
2. **Notes.** What the new version changes.
3. **Install.** Downloads the new version, checks it, and installs it, after a question. The deck restarts.
4. **Check now.** Checks for a new version now.
5. **Beta updates.** On: the deck also gets the betas, the versions with new features for tests. Off: only the stable versions.

Print the parts

The STL files are in lulzbot/stl/, in their print orientation. Load them as they are. The numbers come from slices with the Cura profiles of the mount: LulzBot TAZ Workhorse, SE tool head, 0.5 mm nozzle, 2.85 mm PLA.

	PART	COLOUR	FILAMENT	TIME	COST
	**head** `stl/head.stl`	Navy	87 g	4 h 28 min	$2.00
	**pod** `stl/pod.stl`	Navy	95 g	4 h 42 min	$2.18
	**back** `stl/back.stl`	Orange	69 g	3 h 16 min	$1.59
	**plate** `stl/plate.stl`	Orange	61 g	2 h 18 min	$1.40
	**clip** (2 ×) `stl/clip.stl`	Orange	8 g	0 h 36 min	$0.18
	**grille** `stl/grille.stl`	White	12 g	0 h 42 min	$0.28
	**fan plate** `stl/fan_plate.stl`	White	12 g	0 h 40 min	$0.28

Total: 344 g of PLA (43 m), 16 h 42 min of printing, $7.91 at $22.99 a kilogram. The fit test of the screen opening (stl/fit_display.stl, 22 g, 1 h 10 min) is optional.

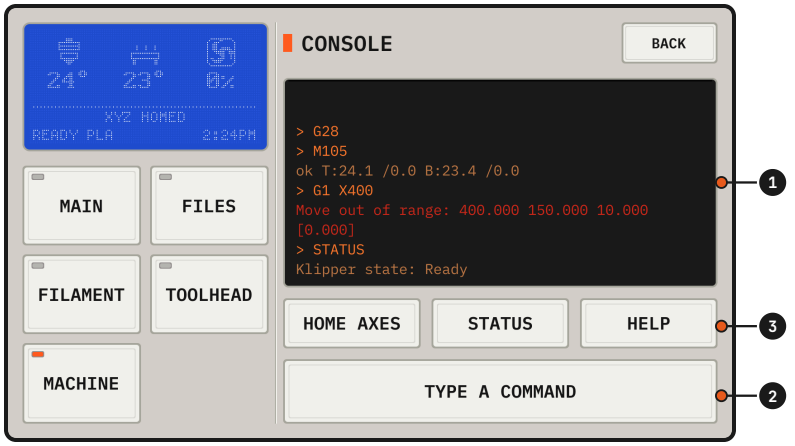
Until you have the metal T-nuts, you can print 2 roll-in nuts (stl/roll_nut.stl, 2 g, 0 h 06 min) and press an M3 hex nut into each one. They roll into the slot at any point on the rail, the same as the metal T-nuts.

Print the fit test first, and check the display in it: the display pocket has 0.3 mm of clearance at each side. Then print the head and the back panel: steps 2 to 4 need them. Then print the plate, the clips, the pod and the grille.

SETTINGS

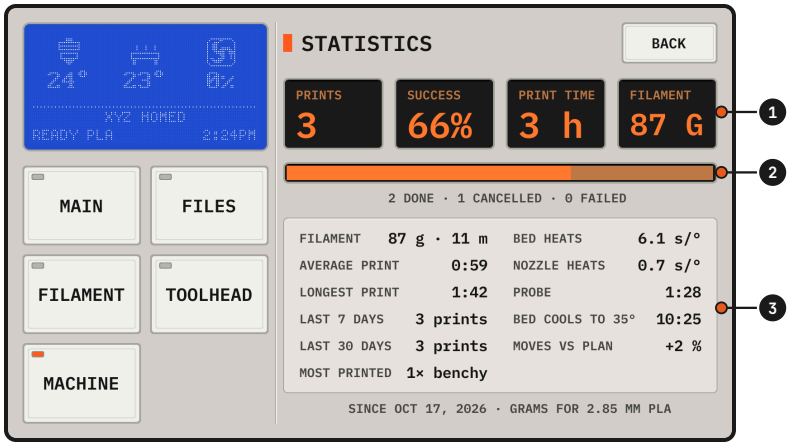
Profiles	One Cura profile for each part: run python3 cura/make_profiles.py --install with Cura closed, then pick “Printdeck - LulzBot ...”.
Printer	LulzBot TAZ Workhorse, SE tool head, 0.5 mm nozzle
Material	PLA, 2.85 mm
Head, pod, grille, fan plate	0.3 mm layers, 1.5 mm walls, 1.2 mm top and bottom, 15 % grid infill, skirt
Back panel, plate	0.4 mm layers (a 0.3 mm first layer), 1.5 mm walls, 15 % grid infill, skirt
Clip, roll-in nut	0.3 mm layers, 100 % infill, with a brim.
Speeds	Infill 75 mm/s (60 mm/s with 0.4 mm layers), inner walls 60 mm/s, outer walls 35 mm/s (30 mm/s on a face that shows), travel 200 mm/s. The flow stays at 12 mm^3/s or less.
Supports	Off for all parts. Every hole, pocket, and hook has 45° faces. A slice of each part with supports on (60°, everywhere) adds 0 g of support.

The console



1. **Replies.** Your commands (amber, after >), the replies of Klipper, and errors (red). Drag down to read older lines.
2. **Type a command.** Opens the keyboard. It shows your last command, so you can change it and send it again.
3. **Quick keys.** Home (G28), Status (STATUS) and Help (HELP, the list of the commands).

Statistics



1. **Totals.** The number of prints, the success rate, the print time and the filament.
2. **Results.** Done (amber), cancelled and failed (red).
3. **Details.** The filament in grams (for 1.75 mm PLA), the average and the longest print, the prints of the last 7 and 30 days, and the file that you print most.

What you need

HARDWARE

	ITEM	WHERE
6 ×	M3 × 10 socket cap head	2 × plate on the cover (upper pair of PP-FP0164); 4 × back panel into the head
2 ×	M3 × 12 socket cap head	2 × plate flange into the back panel
12 ×	M3 × 6 socket cap head	4 × display board (it cuts its thread); 4 × Pi (it cuts its thread); 4 × drive on the back panel
4 ×	M3 × 8 socket cap head	2 × pod onto the plate; 2 × rail clip into its T-nut
2 ×	M2.5 self-tapping round head Phillips, 8 to 11 mm long	2 × fan onto the fan plate (it cuts its thread)
2 ×	M3 hex nut, 5.5 mm across the flats	2 × printed roll-in nuts, in place of the T-nuts (option)
2 ×	M3 roll-in T-nut with spring ball, for 2020 extrusion, 6 mm slot	2 × rail clips
2 ×	M3 washer	2 × under the cover screws
2 ×	M3 × 5 heat-set insert	2 × plate, pod screw blocks (3 to 5 mm fits)
6 ×	M3 × 7 heat-set insert	4 × head, back screws (3 to 7 mm fits); 2 × back panel, flange screws (3 to 7 mm fits)

All the screws are M3 socket cap screws, for a 2.5 mm hex key. A shorter or a longer heat-set insert fits, up to the length in brackets.

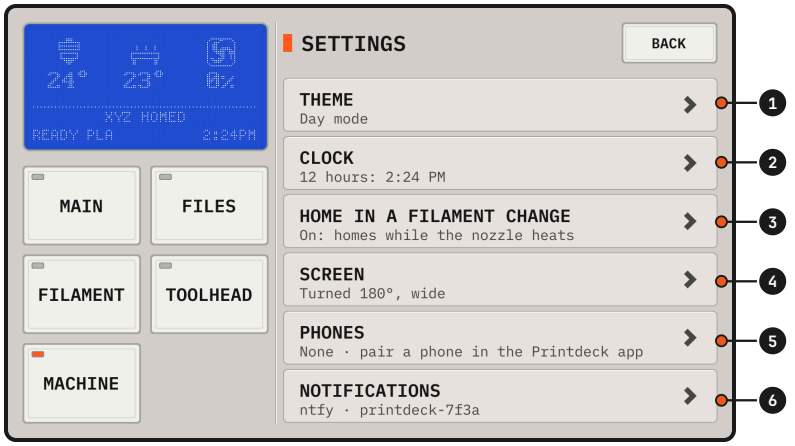
ELECTRONICS AND CABLES

- 1 × Raspberry Pi 5 (from the Slate build)
- 1 × 30 × 30 × 10 mm 5 V fan with the Pi 5 fan plug (4-pin JST-SH), the same as the Slate. From the fan to the fan socket of the Pi, the cable route is 4 cm. The cable must be 8 cm or more, so that the fan plate with the fan comes out of the pod on its cable.
- 1 × Stick-on heatsinks for the Pi 5 (a set), max. 13 mm high. The heatsink of the SoC can be up to 20 × 20 mm.
- 1 × Adafruit 7 inch HDMI display with touch, product 2407 (from the Slate build)
- 1 × 2.5 inch SATA drive with its USB adapter board, the boot drive (from the Slate build)
- 1 × HDMI cable: micro HDMI to HDMI, straight plugs, a flat ribbon of 30 cm or more (the route is 27 cm). The ribbon kit of the Slate fits.
- 1 × Micro USB to USB-A cable, straight plugs, max. 3.5 mm thick, 15 cm or more (the route is 13 cm).
- 1 × USB 3 cable for the drive: micro-B SuperSpeed to USB-A, 15 to 20 cm (the route is 15 cm).
- 1 × USB-A to USB-B cable for the printer: 31 cm in the pod, plus the run behind the printer to the USB socket of the printer board.
- 1 × USB-C power cable: 9 cm in the pod, plus the run behind the printer to the Raspberry Pi 27 W USB-C power supply.

TOOLS

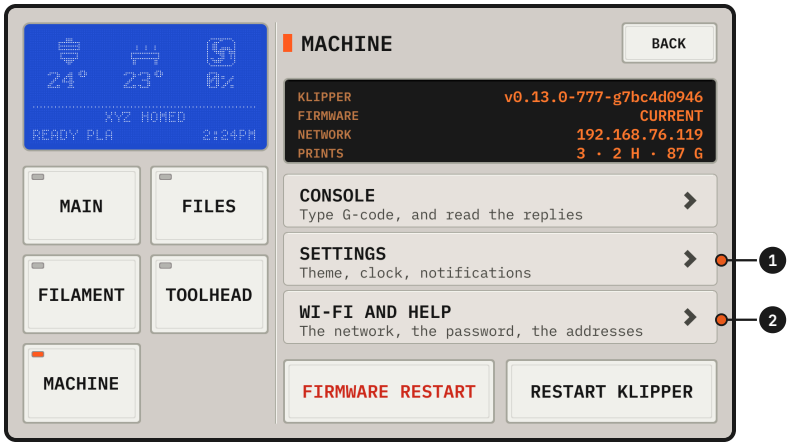
- Drill with a 3.2 mm bit (for the four holes of the Pi)
- Soldering iron with a heat-set insert tip, at 220 °C
- 2.5 mm hex key (M3 socket cap screws)
- One strip of 2 mm foam tape, between the panel and the display board

The Settings page



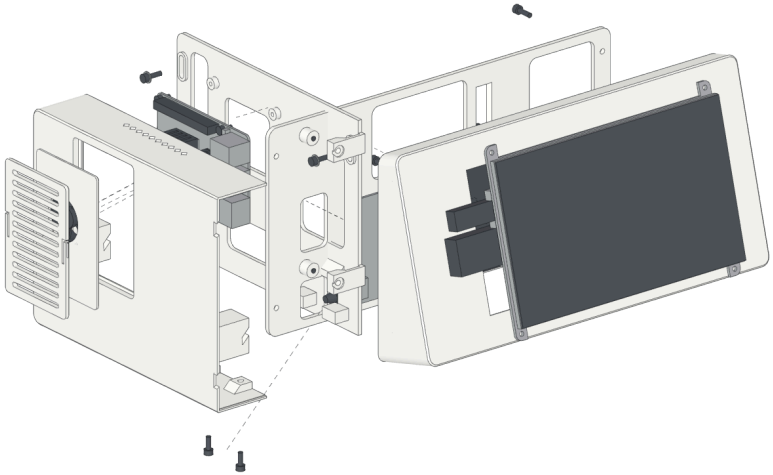
1. **Theme.** The colors of the screen: day, night, or auto (night from 20:00 to 7:00). Tap it to change them.
2. **Clock.** The clock of the screens and the display of the printer: 12 hours (2:24 PM) or 24 hours (14:24). Tap it to change it.
3. **Home in a filament change.** On: the printer homes all axes while the nozzle heats for a filament change. A change during a paused print does not home. Tap it to turn it on or off.
4. **Screen.** The turn of the screen: 0° or 180° for a display that is mounted wide, 90° or 270° for a display that is mounted tall. A tap asks to turn it a quarter turn more, then the deck restarts in the new turn. The touch turns with the screen. A print goes on during the restart. This setting wins over display_rotate in printdeck.txt.
5. **Phones.** The phones with the Printdeck app that get the push messages of the prints. The app pairs with the deck over the network, with the address and the API key. The messages are encrypted from the deck to the phone. Tap the row for the list; a tap on a phone asks to remove it.
6. **Notifications.** Push messages to your phone and watch: print done, print failed, a pause, the end of the first layer, and a roll that runs short. Install the ntfy app on the phone, and subscribe to a topic with a name that is hard to guess. Type the same topic here, or the address of a topic on your own ntfy server. A test message goes out when you save it. An empty topic turns the messages off.

The Machine page, further down



1. **Settings.** The theme, the clock, and the homing in a filament change. See below.
2. **Wi-Fi and help.** The network, the password and the addresses.

Before you start



The mount uses two screws of the printer and the front rail. The upper pair of screws of the side cover (PP-FP0164) holds the plate. Two printed clips on the front slot of the front left rail hold the right end of the head: a metal T-nut rolls into the slot behind the lips, and an M3 × 8 screw pulls it up. Two screws through a flange of the plate hold up the head, and the pod slides onto two hooks of the plate.

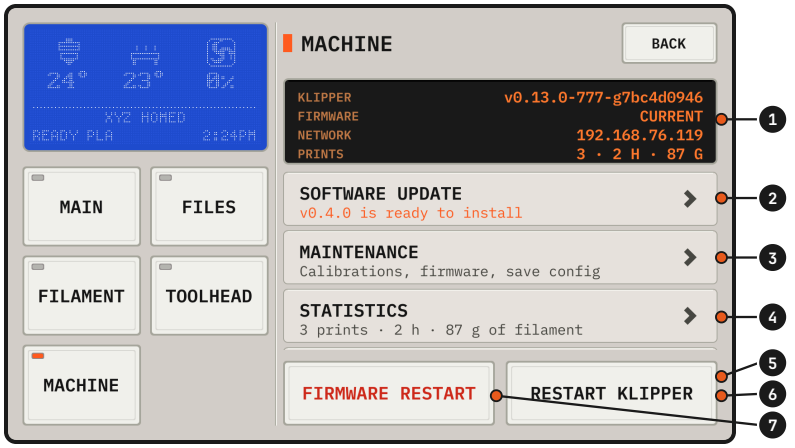
CHECK ON YOUR PRINTER

1. Check the rear screw of the upper pair on the cover: 237.3 mm above the bottom edge of the cover, and about 3.6 mm from the rear edge of the cover. The design uses the value of the LulzBot drawing.
2. Measure the TAZ Workhorse logo on the front of the chassis. The top of the head is 260 mm above the bottom of the chassis; the LulzBot drawing gives the logo at 264 mm, and the drawing is not to scale.
3. Look at the front slot of the front left rail. The clips need a free slot from 150 to 230 mm above the bottom of the chassis: remove the dust cover there, if the rail has one.
4. Move the bed to the front and the X axis to the bottom. Check that the head is clear of the bed, the print, and the tool head: it is 72 to 324 mm to the right of the left face of the chassis, and 125 to 260 mm above the bottom of the chassis.
5. Check behind the printer: the printer USB cable and the power cable leave the pod at the rear face of the chassis, 128 to 149 mm above its bottom.

Look after the printer

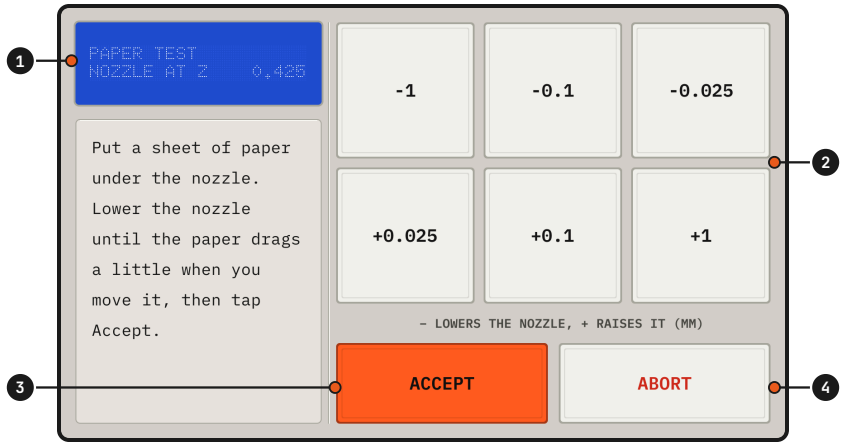
Machine has the information of the printer, the history of the prints, the macros, the console and the settings.

The Machine page



1. **Information.** The version of Klipper, the state of the firmware of the board, the network address, and the totals.
2. **Software update.** The version of the deck, and a new version when there is one. See Update the Printdeck.
3. **Maintenance.** The calibrations, the firmware, and Save config. See Maintenance.
4. **Statistics.** The totals of all prints.
5. **Print history.** Each print, with its result. Tap a print to see its model, its time and its filament, and to print it again.
6. **Restart Klipper.** Restarts the Klipper host, which reads printer.cfg again.
7. **Firmware restart.** Restarts the firmware of the board, for example after a stop.

The paper test



1. **Height.** The height of the nozzle now.
2. **Steps.** Moves the nozzle down (−) or up (+), in mm.
3. **Accept.** Takes the height. Save config then writes it.
4. **Abort.** Stops the test with no change.

> This screen shows by itself while Klipper runs a paper test, also when it started from a macro.

Prepare the drive

The Printdeck starts from the 2.5 inch drive. Write the Printdeck image to it before you build the mount: the drive goes into the head in step 3.

1. Build the image on a computer with Docker: run os/build.sh in the Printdeck repository. It writes os/out/printdeck.img.xz.
2. Connect the drive to the computer with its USB adapter. Write printdeck.img.xz to the drive with Raspberry Pi Imager (Use custom) or balenaEtcher.
3. Open the PRINTDECK partition of the drive, and open printdeck.txt in a text editor.
4. Set display_rotate=180. The display is upside down in the head, so the deck turns its picture. A width larger than the height gives the wide layout.
5. Fill in the name and the password of your Wi-Fi network, if you want to skip the Wi-Fi setup. The deck reads the file at each start, and then clears the password in it.

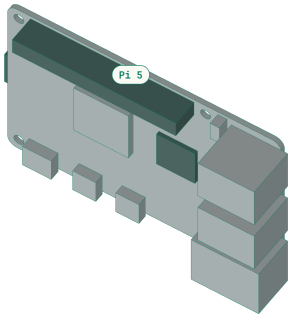
> If the picture is upside down at the first start, set display_rotate=0.

Build the mount

Each step names the parts and the hardware that it needs. The parts from an earlier step have the number of that step.

1 Drill the Pi

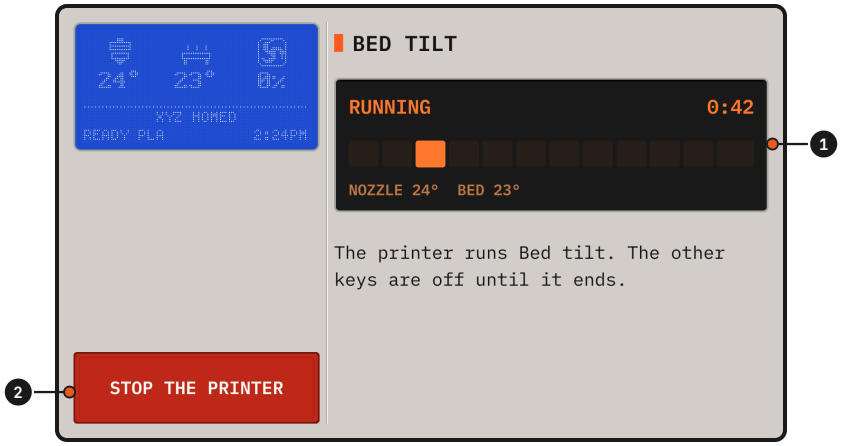
Pi 5 · 3.2 mm drill



1. The mounting holes of the Pi 5 are 2.7 mm, for M2.5. Drill the four holes to 3.2 mm for M3.
2. Drill slowly, from the top of the board, with the board flat on a piece of wood.
3. Keep the drill straight. Stay inside the plated ring of each hole: do not touch the tracks or the parts around it.
4. Remove all the dust from the board with a brush or with air before you continue.

> Skip this step if your Pi already has 3.2 mm holes.

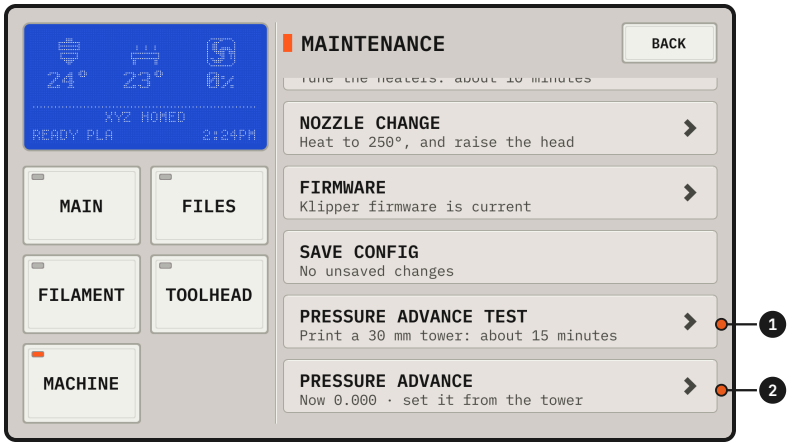
An action runs



1. **Action.** The time since the start, a bar that moves while the action runs, and the temperatures.
2. **Stop the printer.** Stops the printer at once, after one question. Then use Firmware restart.

> While an action runs, the deck shows it on the deck and in each browser, and the other keys are off. The display of the printer shows the name of the action and a moving bar. When the action ends, a short message says that it is done. When it fails, the deck shows the error of Klipper until you tap OK.

The Maintenance page, further down

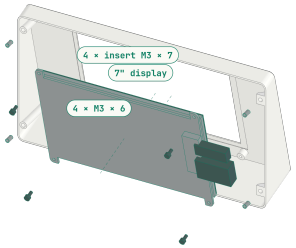


1. **Pressure advance test.** Prints a square tower, 30 mm high, with thin walls. The pressure advance goes up with the height: 0.005 for each mm. Look at the corners: low on the tower they bulge, high on the tower they are rounded. Measure the height where they are sharpest.

2. **Pressure advance.** Type the height of the sharpest corners. The deck sets the pressure advance from it, and writes it in printer.cfg.

2 *Display into the head*

`head` `display` `foam tape` `4 × M3 × 7 insert` `4 × M3 × 6`

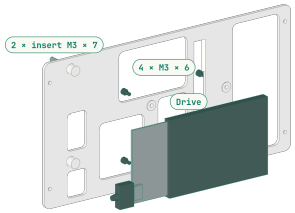


1. Press the four M3 × 7 inserts into the two columns and the two blocks at the back of the head. Use a soldering iron at 220 °C. Stop when the insert is flush.
2. Put the head face down on a soft cloth.
3. Put the display glass down into the pocket: the sockets of the board at the left (seen from the front), the FPC edge of the panel at the top.
4. Put the foam tape on the back of the panel. Put the board on the four bosses, and attach it with four M3 × 6 screws. They cut their own thread in the bosses: turn them until they are snug, and no more.
5. Connect the HDMI cable and the micro USB cable to the board. Both plugs point to the left.

> The screen opening is at the centre of the face: the border is the same at the left and the right.

3 Drive on the back panel

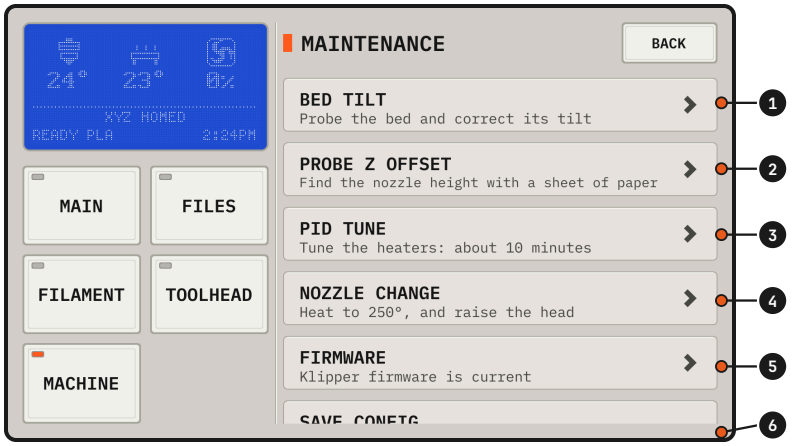
back panel · drive · 2 × M3 × 7 insert · 4 × M3 × 6



1. Press the two M3 × 7 inserts into the two bosses on the front face of the back panel, from the rear.
2. Put the drive on the four pads on the front face of the back panel: the adapter toward the left (the pod side), the M3 holes on the pads.
3. Attach it with four M3 × 6 screws from the rear face of the panel.
4. Connect the micro-B plug of the drive cable to the adapter.

> Press each insert in until it is flush with the rear face, or a little below it. Cut off any ring of melted plastic around the insert. The plate flange has a relief for a small ring, but a large one holds the flange off the back panel, and the two screws then bend the panel at its corner.

The Maintenance page



1. **Bed tilt.** Probes the bed, and corrects its tilt (BED_TILT_CALIBRATE). A printer with a bed mesh shows Bed mesh.
2. **Probe Z offset.** The paper test for the height of the nozzle (PROBE_CALIBRATE).
3. **PID tune.** Heats the nozzle to 210° and the bed to 60°, and finds the PID settings of each heater (PID_CALIBRATE). Save config then writes them.
4. **Nozzle change.** Heats the nozzle to 250° and raises the head, to change the nozzle.
5. **Firmware.** Writes the Klipper firmware of this Printdeck to the board of the printer. Orange text shows old firmware. See below.
6. **Save config.** Writes the results of the calibrations into printer.cfg. Klipper restarts. Orange text shows unsaved results.

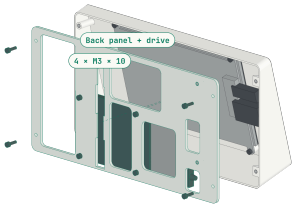
Level and calibrate

Maintenance lists the calibrations that your printer has. The list comes from the config of the printer. Open it from Machine.

1. Open Machine, then Maintenance.
2. Tap Bed tilt or Bed mesh to probe the bed. The deck shows the action until it ends.
3. Tap Probe Z offset for the paper test. Move the nozzle down until a sheet of paper drags a little, then tap Accept.
4. Tap PID tune after you change a heater or a fan. A PID tune runs for about 10 minutes.
5. Tap Save config to write the results. Klipper restarts.
6. Tap Pressure advance test to print the tower. Then tap Pressure advance, and type the height of its sharpest corners.

4 Back panel onto the head

`head (step 2)` `back panel (step 3)` `4 × M3 × 10`

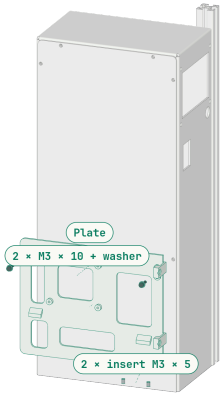


1. Put the display cables through the upper window at the left of the back panel, and the drive cable through the lower window.
2. Put the back panel on the head. Attach it with four M3 × 10 screws from the rear face, into the inserts.

> The heads of the screws stay on the rear face. They are behind the head, and they do not show.

5 *Plate on the cover*

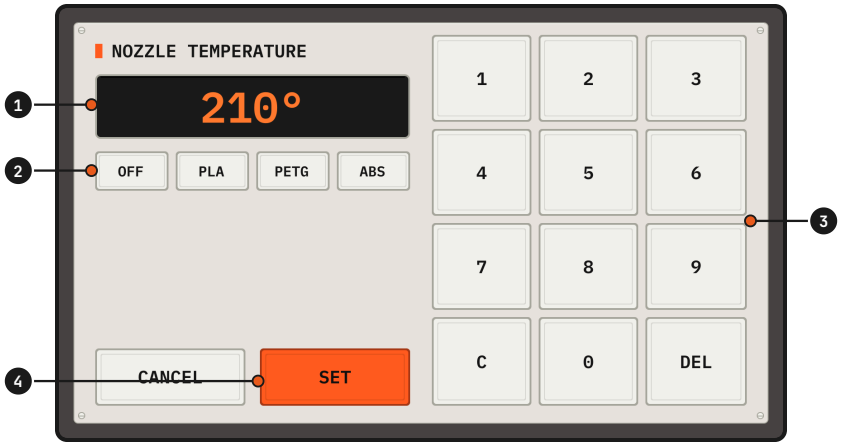
`plate` `2 × M3 × 5 insert` `2 × M3 × 10` `2 × M3 washer`



1. Press the two M3 × 5 inserts into the two blocks at the bottom of the plate (from below).
2. The cover is PP-FP0164, with 10 screws. Remove its upper pair of screws from the left side of the chassis: the front one and the rear one, 237.3 mm above the bottom edge of the cover.
3. Put the plate on the cover, with the slots of the two pads over the two holes. Attach it with two M3 × 10 screws, each with a washer. Leave them loose: the flange screws pull the plate to the back panel first.

> The slots in the pads let the plate fit 5 mm up or down and 1 mm to the front or the rear. The pad is 4 mm thick, so the M3 × 10 goes 3.5 mm into the nut of the chassis (the old screw went 4.0 mm).

The keypad of a temperature

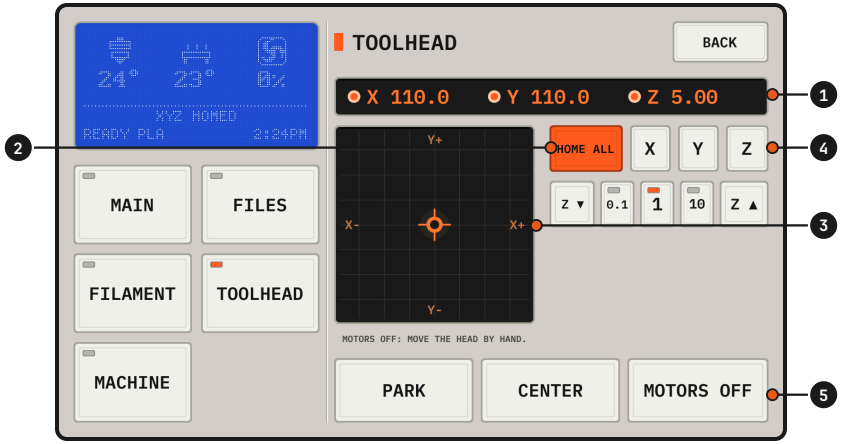


1. **Value.** The temperature that you type.
2. **Presets.** Off, and the temperatures of the materials.
3. **Keys.** The digits. C clears the value, DEL removes the last digit.
4. **Set.** Sets the heater. Cancel closes the keypad with no change.

Move the head and heat

1. Open Toolhead, and tap Home all after each start of the printer.
2. Hold the joystick to move the head. Use Z for the height.
3. Tap a temperature on the idle screen to set it with the keypad.

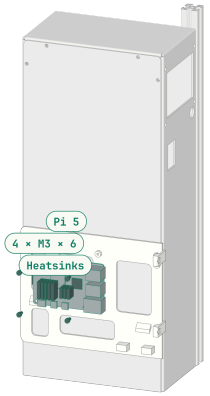
The Toolhead page



1. **Position.** X, Y and Z of the head. A lit LED shows a homed axis.
2. **Home.** Homes all axes, or one axis. Home after each start of the printer.
3. **Joystick.** Hold a finger on the pad to move the head in X and Y. The farther from the middle, the faster it moves.
4. **Z.** Moves Z up or down by the step: 0.1, 1 or 10 mm.
5. **Park, Center, Motors off.** Park moves the head to the back left. Center moves it to the middle of the bed. Motors off lets you move the head by hand. During a pause, Home, Center and Motors off are off, so that the print can continue. Park lifts the nozzle 10 mm before it moves.

6 Pi on the plate

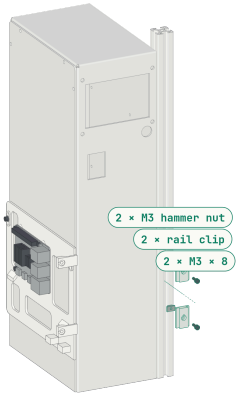
Pi 5 (step 1) · heatsinks · 4 × M3 × 6



1. Stick the heatsinks on the Pi: on the SoC (the large chip at the middle) and on the RP1 (the chip between the SoC and the USB ports). A heatsink can be up to 13 mm high.
2. Put the Pi on the four bosses: the USB ports toward the front of the printer, the USB-C and HDMI ports down.
3. Attach it with four M3 × 6 screws. They cut their own thread in the bosses: turn them until they are snug, and no more.

7 Clips in the rail

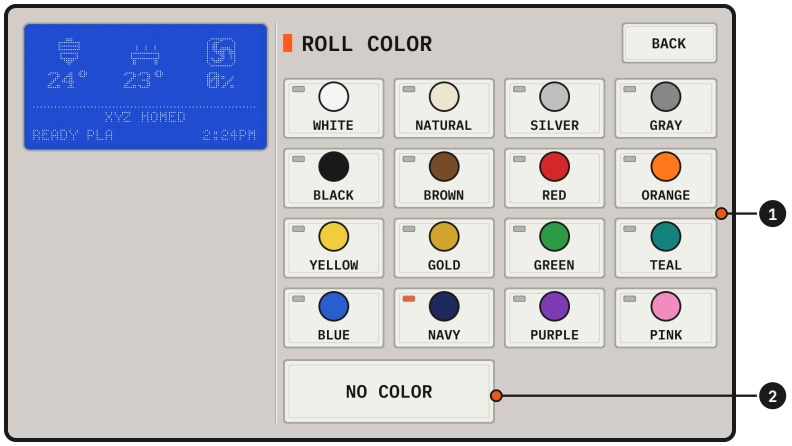
2 × rail clip | 2 × M3 roll-in T-nut | 2 × M3 × 8



1. Hold the back panel on the front face of the front left rail, with its bottom edge 125 mm above the bottom of the chassis. Mark the top of the two long slots in the panel on the rail.
2. At each mark, put one edge of a T-nut into the front slot of the rail, with the thread toward you. Roll the nut in until it is behind the lips. The spring ball holds it in the slot.
3. Put a clip on the rail at a mark, with its key along the rail. Put an M3 × 8 screw through the counterbore in the key, and turn it into the nut.
4. Tighten the screw until the clip is tight on the rail. Do the same with the second clip.

You can roll the nut into the slot at any point on the rail. You do not remove the ends of the rail. If the nut moves away from the mark, push it back with a small screwdriver. Until you have the metal T-nuts, you can use the printed roll-in nuts (stl/roll_nut.stl). Press an M3 hex nut into the pocket of each one with pliers. Put one long edge into the slot, with the hex nut away from you, and roll the nut in behind the lips. Tighten the screw gently: the printed nut is not as strong as a metal T-nut.

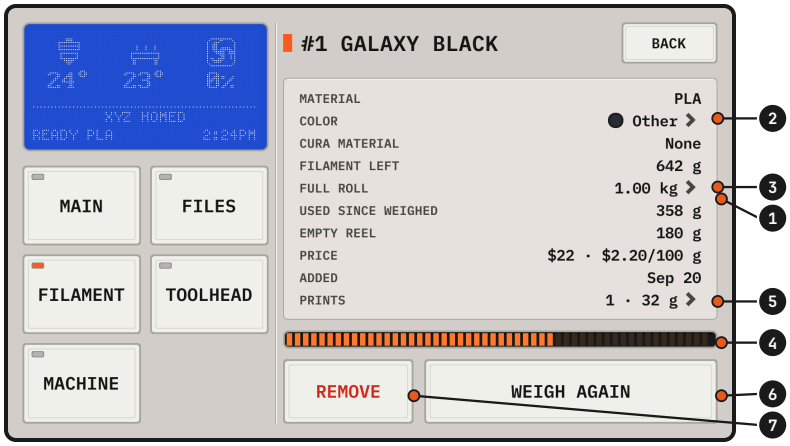
The color of a roll



1. **Colors.** The colors of filament. The color of the roll now is lit.
2. **No color.** Removes the color.

A tap on a color sets it, and closes the page. The Cura plugin gives the material of the roll in Cura the same color. A color that you set in Cura comes back to the deck.

One roll

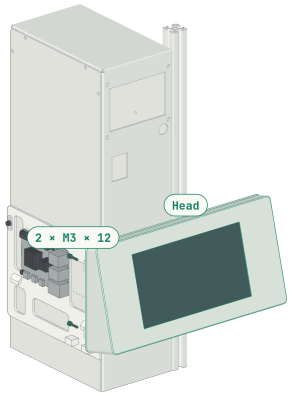


1. **Details.** The filament on the roll, the filament that prints used, and the price.
2. **Color.** The color of the roll. Tap it to choose a color. The print screen and Cura show it.
3. **Full roll.** The filament of a full roll as it is sold: 1 kg unless you change it. Tap it to type it, for example 250 g or 2 kg. The bar and the price per gram use it, so a roll that you added part used shows how much of a full roll is left.
4. **Left.** The filament on the roll, as a part of a full roll.
5. **Prints.** The number of prints that used the roll, and their filament. Tap it for the list of the prints. A tap on a print opens its page.
6. **Weigh again.** Sets the filament from the weight on a scale.
7. **Remove.** Removes an empty roll from the list.

Each roll has a number, for example #3. The deck does not use a number again. For a label with a QR code, open printdeck.local/rolls/labels in a browser, and print the sheet on round labels of 2 inches (Avery 22807). Choose the first free label of the sheet, so a sheet lasts for many rolls. A phone camera on the label opens the page of the roll: Load opens the filament change on the deck, with the roll.

8 Hang the head

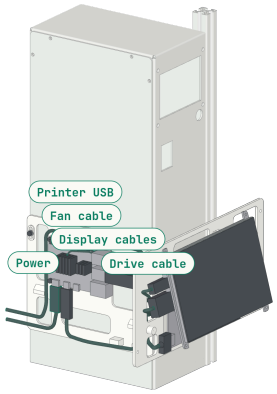
`head (step 4)` `2 × M3 × 12`



1. Hold the head 30 mm above its place, with the two windows of the back panel over the two clip keys. Push the back panel against the clips.
2. Let the head drop 30 mm. The keys go into the dovetail slots.
3. Lead the display cables and the drive cable out of the windows into the pod.
4. From inside the pod, put the two M3 × 12 screws through the flange of the plate into the back panel. Tighten them, then tighten the two cover screws of the plate.

The two screws through the flange hold up the head. The clips hold its right end against the rail.

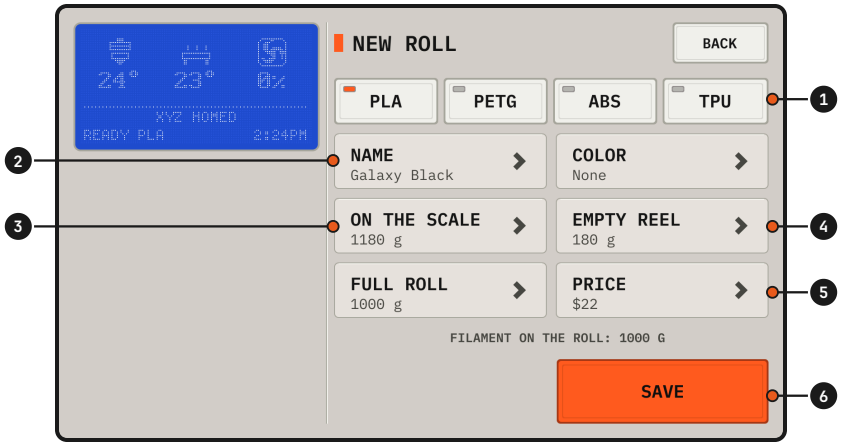
9 Cables



1. Connect the HDMI cable to the HDMI 0 port (next to the USB-C port). Push the plug straight in. Let the cable go straight down from the plug, then turn it to the front in a wide curve along the bottom of the pod. Do not bend it at the plug.
2. Connect the touch cable to the upper USB 3 port (blue), and the drive cable to the lower USB 3 port.
3. Connect the printer cable to the lower USB 2 port. Lead it up the front of the pod, over the Pi, down at the rear end of the Pi, and out of the upper notch in the rear wall.
4. Connect the power cable to the USB-C port. Lead it down and to the rear, out of the lower notch.
5. Behind the printer, connect the printer cable to the USB socket of the printer board, and the power cable to the power supply.

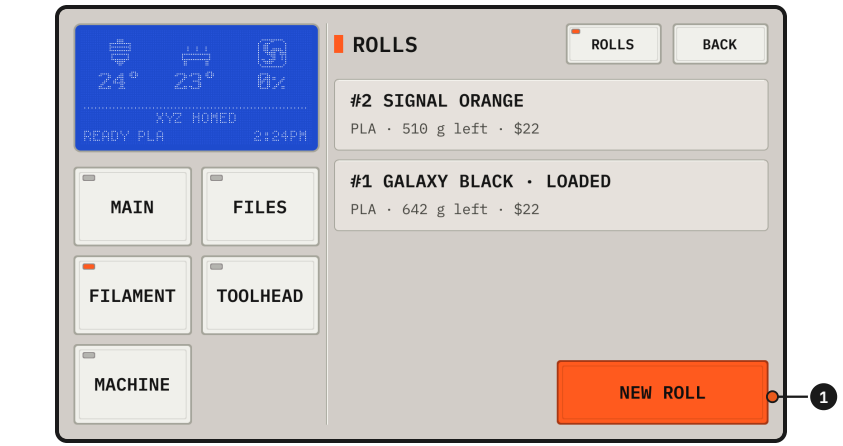
> No cable goes through a hole in the printer, and no cable shows on the front or the side.

A new roll



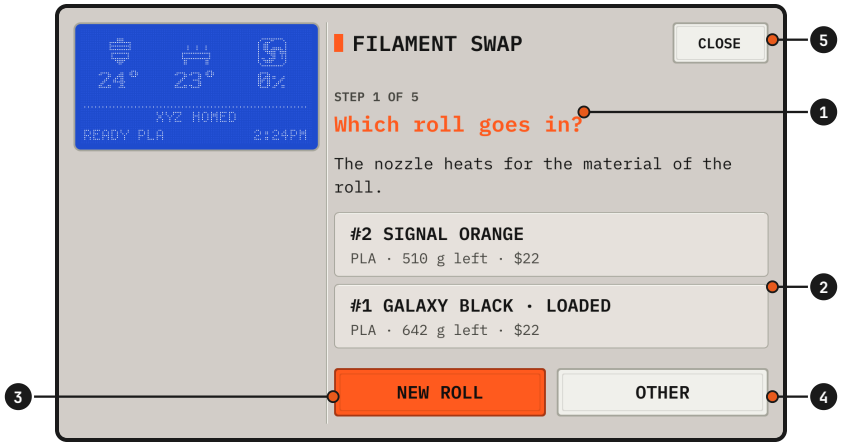
1. **Material.** The material of the roll.
2. **Name.** A name that you type, for example the color and the brand.
3. **On the scale.** The weight of the roll with its reel, in grams.
4. **Empty reel.** The weight of an empty reel of the same brand. The deck keeps the last one.
5. **Price.** The price of the roll in dollars. The deck shows the cost of each print from it.
6. **Save.** Adds the roll and puts it in the printer.

The rolls



1. **New roll.** Adds a roll without a filament change.

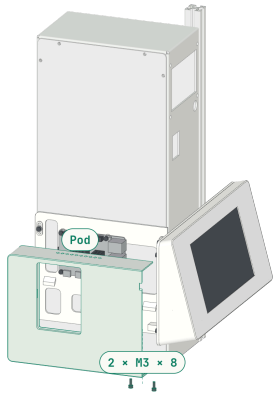
The guided change, step 1



1. **Step.** The step of the change.
2. **Rolls.** Your rolls. Tap the roll that goes in. The nozzle heats for its material.
3. **New roll.** Adds a roll: its material, its name, its weight on a scale, the weight of an empty reel, and its price.
4. **Other.** A filament with no roll: choose the material or type a temperature.
5. **Close.** Stops the change. After the nozzle starts to heat, Close also turns the nozzle off. During a pause, the nozzle goes back to the temperature of the print.

10 *Pod*

pod · 2 × M3 × 8



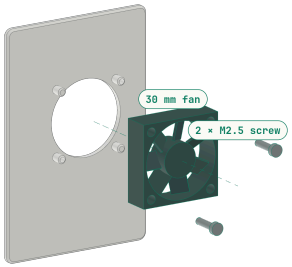
1. Connect the fan cable to the fan socket of the Pi: the small 4-pin socket between the end of the GPIO header and the USB ports. Lay the fan on the Pi.
2. Hold the pod about 6 mm low and 6 mm out from its place, and 4 mm to the rear. Move it 4 mm to the front, then push it up and in at 45° onto the two hooks of the plate. The rear wall goes over the two cables.
3. Put the two M3 × 8 screws up through the bottom of the pod into the plate. They pull the pod up, and the 45° hooks pull it onto the plate.

> Two notches in the front edge of the pod are for the heads of the two screws into the left wall of the head.

11 Fan onto the fan plate

`fan plate` `fan`

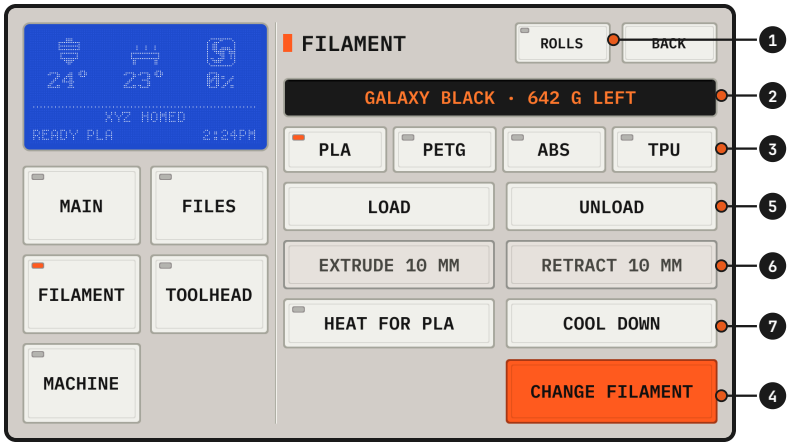
`2 × M2.5 self-tapping round head Phillips, 8 to 11 mm long`



1. Take the fan out of the pod through the grille pocket. Its cable stays on the Pi.
2. Hold the fan plate with its four posts toward you. Turn the fan so that it blows away from the fan plate, onto the Pi. Most fans show the direction of the air with an arrow on the frame.
3. Push the fan onto the four posts, until it is flat on the fan plate. The two solid posts hold it in place.
4. Put the two M2.5 screws into the other two holes, from the far face of the fan. Turn them into the hollow posts until they are snug, and no more. They cut their own thread.

> A screw of 8 to 11 mm fits: it goes through the fan hole into the post. A 10 mm screw has 4 mm of thread. The fan cable leaves the frame from a side: it does not touch the posts or the screw heads.

The Filament page



1. **Rolls.** Opens the list of your rolls. On the list, the key stays in its place with its LED lit, and a tap comes back here.
2. **Loaded.** The roll that is in the printer and the filament on it, or the material and its temperatures.
3. **Material.** Tap the material that is in the printer. The deck asks before it changes the loaded material.
4. **Change filament.** The guided change: choose, heat, unload, insert, load, check. The printer homes while the nozzle heats. Turn this off on the Settings page.
5. **Load, Unload.** Only the load, or only the unload.
6. **Extrude, Retract.** Pushes or pulls 10 mm. The nozzle must be at 170° or more.
7. **Heat, Cool down.** Heats the nozzle for the loaded filament, or turns the heaters off.

Change the filament

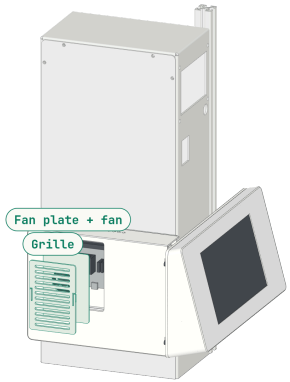
Set the loaded filament here, so that Preheat uses its temperatures. The guided change heats the nozzle, unloads the old filament and loads the new one.

The deck keeps a list of your rolls of filament. Weigh a new roll with its reel, and the deck subtracts the weight of the empty reel. Each print subtracts the filament that it used from the roll in the printer.

1. Open Filament, then tap Change filament.
2. Tap the roll that goes in, or tap New roll and weigh it. The nozzle heats for its material.
3. Tap Unload, and hold the filament lightly while it comes out.
4. Push the new filament into the extruder until it stops, then tap Load.
5. Check that clean plastic of the new color comes out, then tap Done.

12 Fan plate and grille

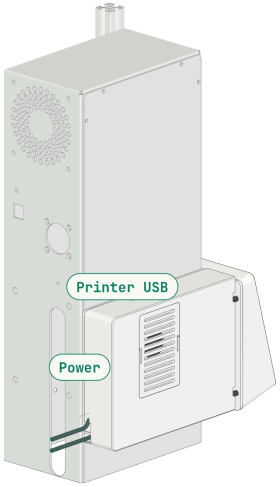
`fan plate with the fan (step 11)` `grille`



1. Push the fan plate with the fan into the grille pocket of the pod from the outside, the fan first. The fan plate goes down onto the ledge of the pocket. The cable stays inside the pod.
2. Hold the grille at the pocket, with its tabs at the sides. Push it into the pocket over the fan plate. Push both sides until the two tabs click. The grille is flush with the pod.

> To get to the fan, put a small screwdriver into a vent slot near a tab, and lift the grille out. Then pull the fan plate out at its air hole. The fan comes out with it, on its cable.

Connect and start



CABLE	LENGTH	ROUTE
HDMI	27 cm	HDMI 0 port to the display, plug to plug
Touch USB	13 cm	upper USB 3 port to the display, plug to plug
Drive	15 cm	lower USB 3 port to the drive, plug to plug
Printer USB	31 cm	lower USB 2 port to the rear face of the chassis
Power	9 cm	USB-C port to the rear face of the chassis

1. Connect the power cable to the Raspberry Pi 27 W USB-C power supply behind the printer.
2. Connect the printer cable to the USB socket of the printer board.
3. Turn on the printer and the power supply. The deck starts from the drive and opens the setup. The guides after this page show the setup.

Lock Screen widgets	The rectangle, the circle and the line of text of the complications.
Live Activity	During a print, the model, the time left, the bar, the end and the layer on the Lock Screen and in the Dynamic Island.
Notifications	Print done, print failed, paused, first layer done, and a roll that runs short.

The iPhone and Apple Watch apps

The Printdeck app for iPhone shows the deck as its touch screen: the display of the printer, the print, the roll and the model. It can pause, resume and cancel a print, and start a file of the deck. The watch app shows the print on a blue dot-matrix display.

The apps talk to the deck on the local network. The watch reaches the deck on Wi-Fi, or through the iPhone when it is near.

1. Open Printdeck on the iPhone. It finds the deck on the network. Tap Connect.
2. Tap Allow on the touch screen of the deck. The app gets its own key, and sends it to the watch.
3. For push messages, open Menu, then Notifications, and tap Pair for notifications. Allow the notifications.
4. To add the complication, edit the watch face and choose Printdeck.
5. To add a widget, touch and hold the Home Screen or the Lock Screen, tap Edit, and choose Printdeck.

iPhone app	The display of the printer, the keys, the roll and the model of the print. Pause, Resume and Cancel during a print. Print a file opens the files of the deck.
Watch app	The progress, the time left and the end. Turn the Digital Crown for the model, the layer times and the filament. Tap for Pause, Resume or Cancel.
Complications	The progress and the time left on the watch face: a rectangle, a ring of dots, a corner and a line of text.
Home Screen widgets	The small widget shows the status of the watch. The medium widget shows the status screen of the deck.

TO OPEN THE MOUNT

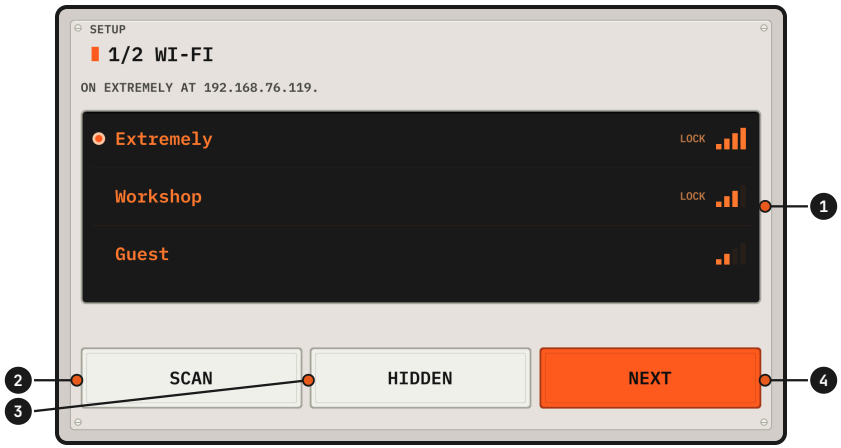
1. Lift the grille out at the fingernail notch, and disconnect the fan cable. Remove the two screws under the pod. Move the pod down and out at 45 degrees, off the hooks, then 4 mm to the rear and out.
2. Disconnect the cables from the Pi. Remove the two flange screws from inside the pod.
3. Lift the head 30 mm and pull it off the clip keys. The display and the drive come off with it.
4. To get to the display or the drive, remove the four screws from the rear of the back panel.

Connect to Wi-Fi

At the first start, the deck makes its password and opens the setup.

1. Switch on the Printdeck. The setup opens.
2. Tap your Wi-Fi network, then type its password.
3. Wait until the deck is connected, then tap Next.
4. Write down the address and the password, then tap Finish.

The setup: step 1



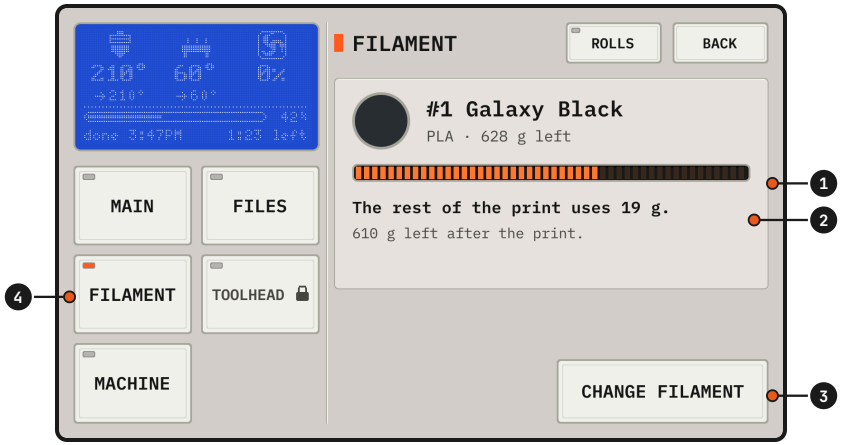
1. **Networks.** The Wi-Fi networks near the deck, with the signal. A lock shows a network with a password. Tap a network, then type its password.
2. **Scan.** Looks for the networks again. The list also scans by itself.
3. **Hidden.** For a network that does not show its name: type the name and the password.
4. **Next.** Goes to the summary when the deck is connected.

Look around during a print

The other screens are read only for the printer while it prints: the files, the filament and the rolls, the history. A tap cannot change the print by mistake. The deck goes back to the print at a pause, at a new print, and after 2 minutes with no touch.

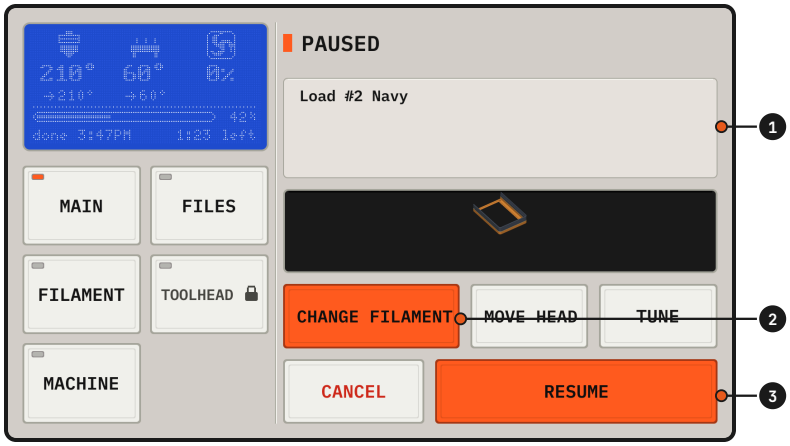
During a print, Main is the print. The key panel opens the other pages as at any time, and Main goes back to the print.

Filament during a print



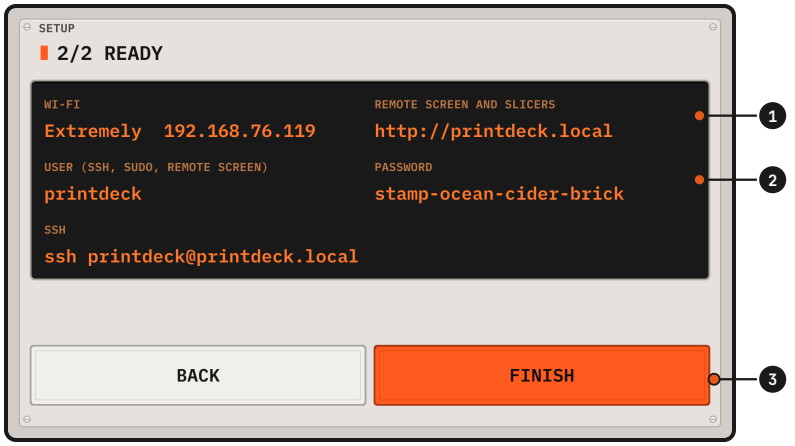
1. **Roll.** The roll in the printer, and the filament that is left on it.
2. **Lasts.** The filament of the rest of the print, and what is left on the roll after it. Red text shows a roll that runs out before the end.
3. **Change filament.** Pauses the print, and opens the filament change. At its end, the nozzle goes back to the temperature of the print: tap Resume.
4. **Pages.** The key panel opens each page, as when the printer does not print.

A color change



1. **Where.** The roll of the next color, from the Colors tool of the Cura plugin.
2. **Change filament.** Opens the filament change with that roll. Unload, load the new filament, and tap Done.
3. **Resume.** Draws a short prime line at the front of the bed, then goes back to the part and continues.

The setup: step 2



1. **Address.** The address of the deck on your network. Open it in a browser for the web pages.
2. **Password.** The password of the web pages. To see it again, open Machine > Wi-Fi and help.
3. **Finish.** Closes the setup. The deck does not open it again.

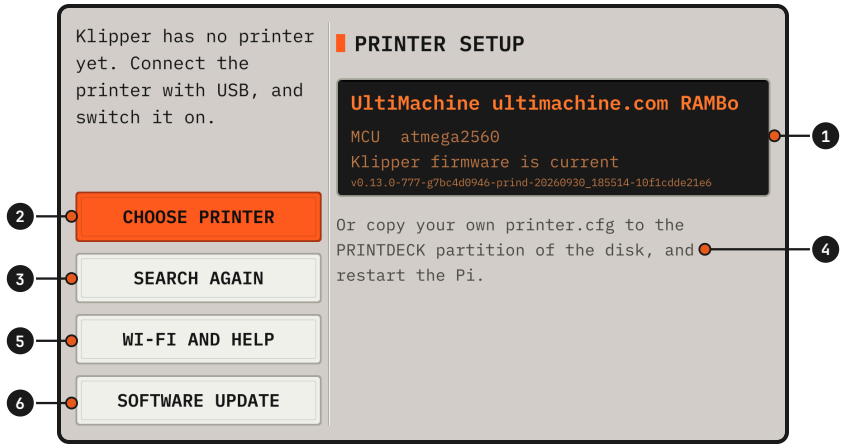
> You can also set the Wi-Fi before the first start: put the disk in a computer, and fill in printdeck.txt on the PRINTDECK partition.

Choose your printer

A new Printdeck has no printer config. The deck then shows the printer setup.

1. Connect the printer to the Printdeck with a USB cable, and switch the printer on.
2. Wait until the deck finds the board. Tap Search again if it does not show.
3. Tap Choose printer. The printers of the brand of your board are at the top.
4. Tap your printer, then Use. Klipper restarts with its config.

No printer yet



1. **Board.** The board that the deck found on USB, its MCU, and its Klipper firmware.
2. **Choose printer.** Opens the list of Klipper's example configs. See below.
3. **Search again.** Looks for the board again, for example after you connect the printer.
4. **Your own config.** You can also copy your own printer.cfg to the PRINTDECK partition, and start the deck again.
5. **Wi-Fi and help.** The network, the address and the password of the deck.
6. **Software update.** Installs a new version of Printdeck OS. A new version can fix the problem.

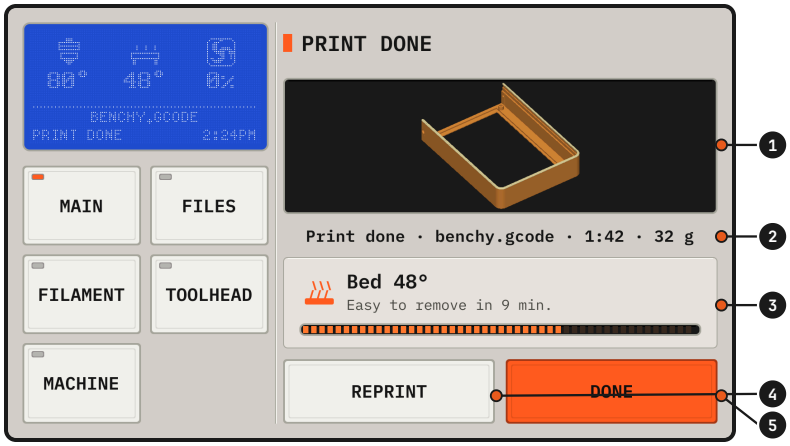
Print in two colors

The Colors tool of the Cura plugin gives a model the roll of another color, for example raised letters on a plate. The print pauses at the first layer of that model, and the deck asks for its roll.

On Resume the deck draws a short line at the front of the bed, behind the prime line of the start. The ooze of the new filament stays on that line. The nozzle then goes back to the part with the filament pulled back a little, and pushes it forward at the first point of the layer. Each change of a print draws its line 2 mm behind the line before it.

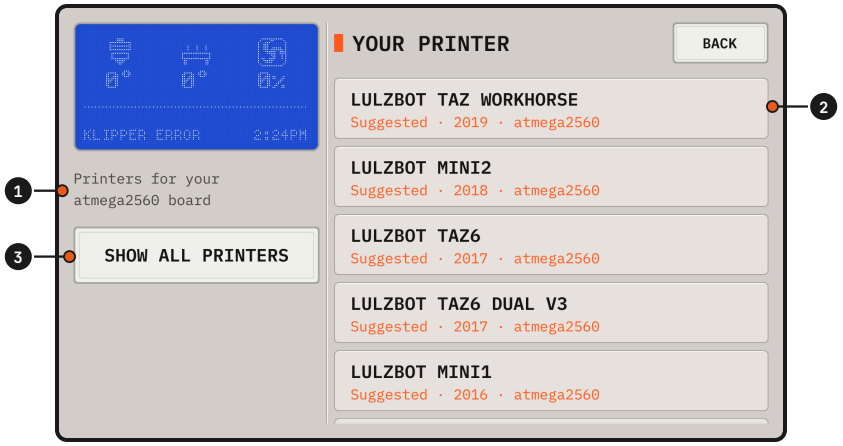
1. In Cura, put the letters on the part. Select the letters, open the Colors tool, and tap their roll.
2. Print. At the first layer of the letters, the printer parks and the deck shows Change filament in orange.
3. Tap Change filament. Unload, load the roll of the letters, and tap Done.
4. Pull the ooze off the nozzle with tweezers, then tap Resume.

Done



1. **Model.** The model of the print.
2. **Result.** Done, Cancelled or Failed, the file, the time and the filament of the print.
3. **Bed.** The temperature of the bed, and the time until the part comes off easily (35°). The deck learns the time from the cooling after each print. The bar is full at 35°.
4. **Reprint.** Prints the file again.
5. **Done.** Closes this screen.

Choose your printer



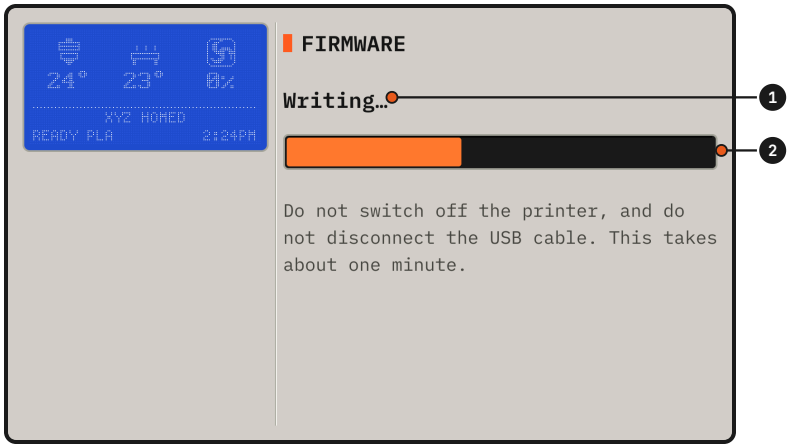
1. **Chip.** The list shows the printers with the same chip as your board.
2. **Printer.** The brand, the model, the year and the chip. Tap your printer. The deck writes its config as printer.cfg, with the port of your board, and restarts Klipper. The old printer.cfg stays as printer.cfg.bak.
3. **Show all printers.** Shows the printers for all chips.

Install the Klipper firmware

The board of the printer must run the Klipper firmware of this Printdeck. The deck writes it through the USB cable.

1. Open Machine, then Maintenance, then tap Firmware. A new board shows Install Klipper firmware on the setup screen.
2. Tap Write. Do not switch off the printer until the deck shows the result.
3. Tap OK. Klipper restarts and connects to the board.

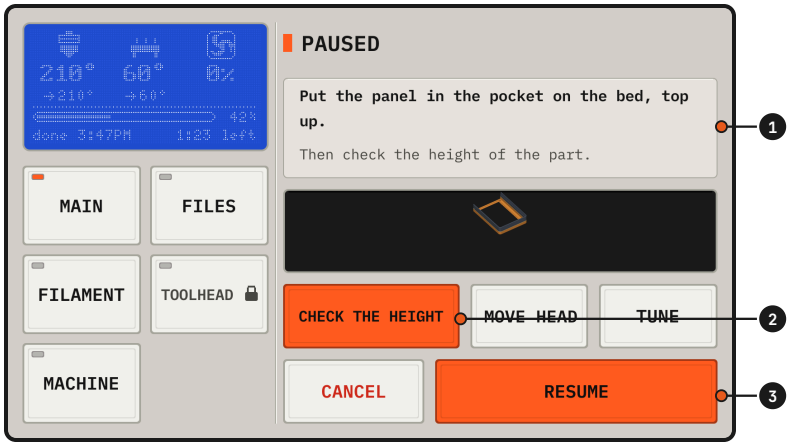
The firmware flash



1. **Step.** Starting the bootloader, Writing, then Checking.
2. **Progress.** The part of the firmware that is done.

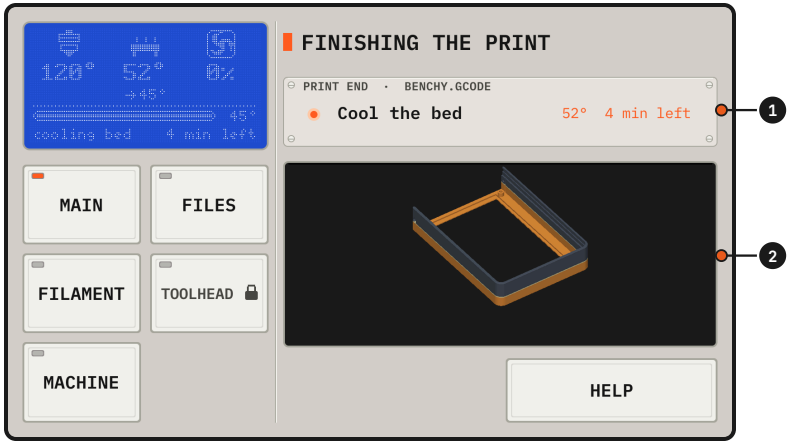
> The deck stops Klipper, resets the board into its bootloader, writes the firmware, and reads it back to check it. Then Klipper starts again. The setup screen and the Klipper error screen also show this key when the board has no Klipper firmware or old firmware. Printdeck can flash boards with an atmega2560 and an STK500v2 bootloader now, for example the RAMBo and RAMPS boards.

A pause of the file



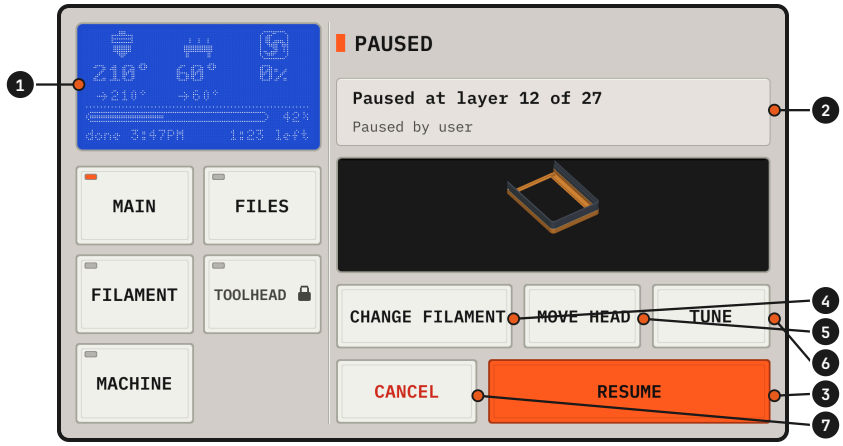
1. **Where.** The message of the file, for example for a part in a pocket on the bed (Print on a part in the Cura plugin).
2. **Check the height.** A paper test on the part. The print goes on from the measured height.
3. **Resume.** Primes the nozzle and continues. With no height check, the deck asks first.

The end of the print



1. **Steps.** The steps of the end macro of your printer, for example the cooling of the bed before the print moves to the front.
2. **Model.** The finished model.

Paused



1. **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.
2. **Where.** The layer of the pause, and the message of the printer.
3. **Resume.** Continues the print.
4. **Change filament.** The guided filament change, for a new color. See Change the filament.
5. **Move head.** Opens Toolhead, to move the head out of the way.
6. **Tune.** Changes the print.
7. **Cancel.** Stops the print, after a question.

The deck

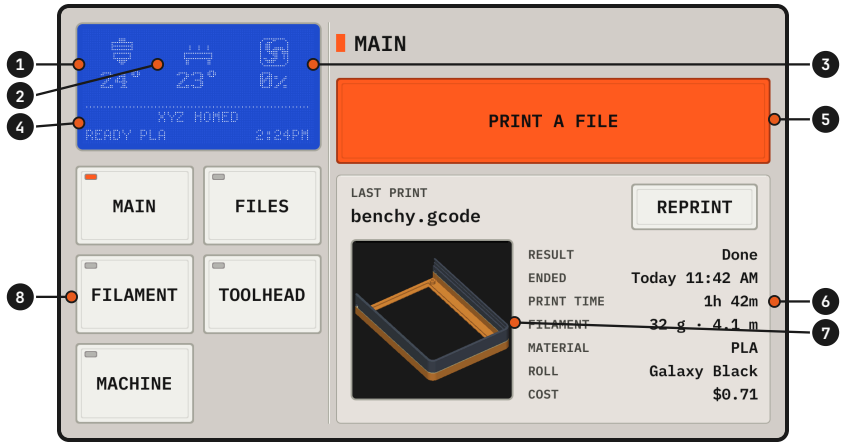
Printdeck is a touch screen for a 3D printer that runs Klipper. It shows the print, and it controls the printer.

When the printer does not print, the deck shows the idle screen. The largest key is the action that you need most: Print a file.

The keys are the keys of a machine. A key with an LED has two states, and its LED is lit while it is on: the page that is open, the material in the printer, Preheat. A key with no LED does an action: an orange ring shows that it took the tap.

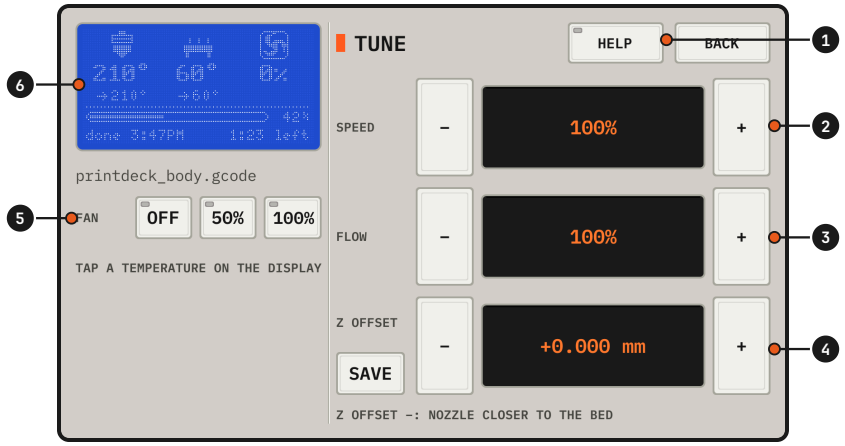
Each screen has the same places. At the left, the display of the printer, and under it the key panel: Main (the idle screen), Files, Filament, Toolhead and Machine. At the right, the page that is open: its title and Back at the top, and its actions in one row at the bottom, with the main action at the right.

The idle screen



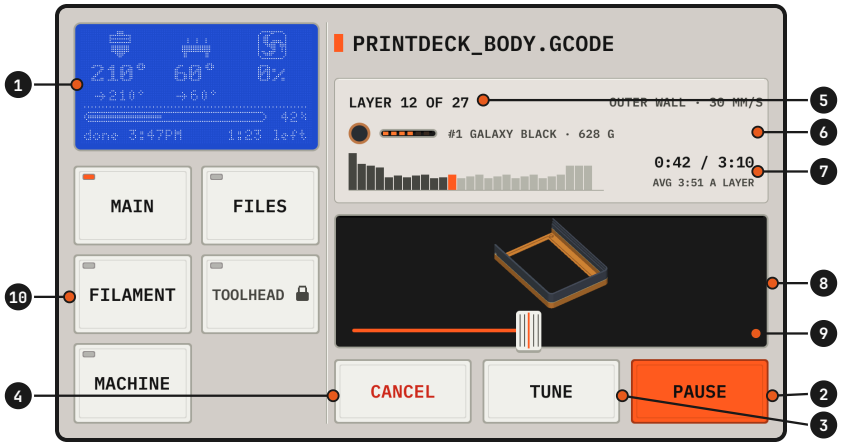
1. **Nozzle.** The nozzle now, and its target under it while the heater is on. The icon is filled while the heater is on, and waves rise while it heats. Tap it to set the temperature.
2. **Bed.** The same for the bed.
3. **Fan.** The part cooling fan. Its blades turn at its speed.
4. **Status.** The position of the head, the state of the printer with the loaded filament, and the clock. This is the status screen of the display of the printer: the same picture on both screens.
5. **Print a file.** Opens the files. See Files.
6. **Last print.** The last print, as on its page of the history: the result, the time, the filament, the roll and the cost. Reprint starts it again, after a question. Preheat and Cool down are in Filament.
7. **Model.** The model of the last print, in the color of its roll. Tap it to open the page of the print in the history.
8. **Pages.** The key panel: Main (this screen), Files, Filament, Toolhead and Machine. The LED of the page that is open is lit. Each page has its own section.

Tune



1. **Help.** Opens Wi-Fi and help: the network, the address and the password of the deck.
2. **Speed.** The speed of the print, in percent of the speed of the slicer.
3. **Flow.** The amount of plastic, in percent.
4. **Z offset.** The height of the nozzle above its normal height. Save writes it into the config of the printer, as on the first layer screen.
5. **Fan.** The part cooling fan.
6. **Display.** Tap the nozzle or the bed on the display to change its temperature.

Printing



1. **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.
2. **Pause.** Pauses the print.
3. **Tune.** Changes the print while it runs. See Tune.
4. **Cancel.** Stops the print, after a question.
5. **Layer.** The layer that prints now, and the kind of the move now with the live speed of the head.
6. **Roll.** The roll in the printer: a dot in its color, the filament that is left on it, its name, and its grams. The filament of the print counts while it prints. Cura gives a roll its color.
7. **Layer times.** A bar for each layer: its time. The layers that are done are dark, with the time that they took. The layer now is orange, and the layers to come are light, with the time of the slicer. At the right, the time of the layer now and its plan, and the mean time of a layer. A short layer can need more cooling.
8. **Model.** The model up to the layer that prints now, in the color of the roll.
9. **Layer slider.** Drag it to look at another layer. Live goes back to the layer that prints.
10. **Pages.** During a print, Main is the print. The other keys open their pages while the print goes on, and Main comes back to the print.

The display of the printer

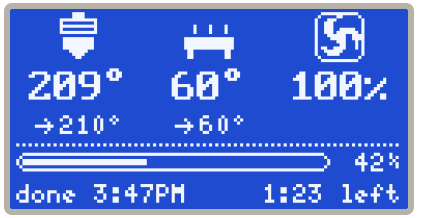
When the printer has a display, the deck draws it. The display shows the status screen, the same picture as the idle screen of the deck.

Turn or click the knob to open the menu. The menu has the same pages and keys as the touch screen: the knob moves the cursor, and a click taps the key. The touch screen shows the cursor with an orange ring. The top item goes back: Status on the home menu, Back on a page. A long click also goes back. With no turn for 20 seconds, the display shows the status screen again.

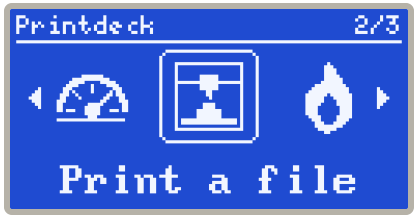
While an action runs, the display shows an animation.



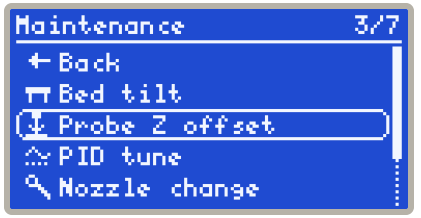
THE STATUS SCREEN



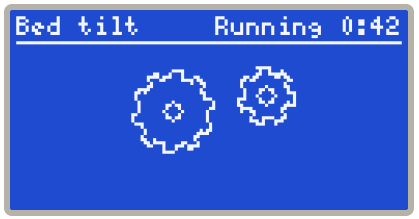
DURING A PRINT



THE HOME MENU: TURN THE KNOB TO CHOOSE



A PAGE: MAINTENANCE



AN ACTION RUNS

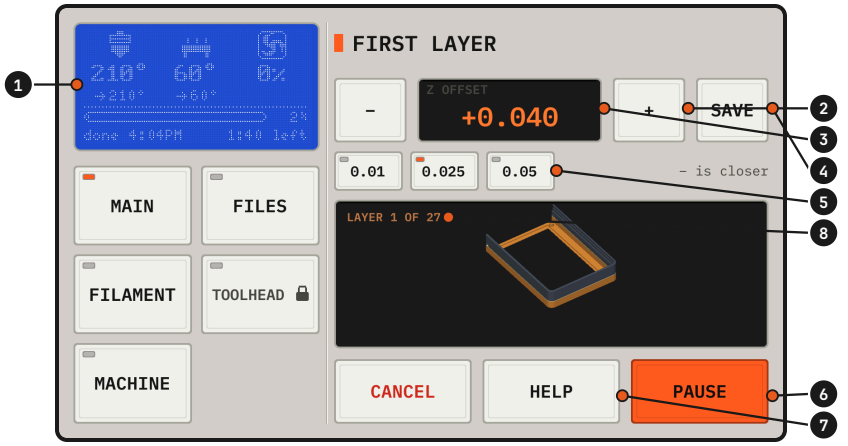
The deck in a browser

Open http://printdeck.local in a browser on the same network, and log in with the password of the deck. The browser shows the same screens as the touch screen, sharp at the pixel density of the phone. Each browser has its own screens, so a phone can show the files while the deck shows the print.

On a phone, use Add to Home Screen: the deck then opens as an app, with no browser bars.

The page http://printdeck.local/system shows the version, the services, the logs and the API key for slicers. It also installs an update: choose the update file, then Install. The deck writes the other system slot and restarts into it. If the new system does not start, the deck goes back to the old one.

The first layer



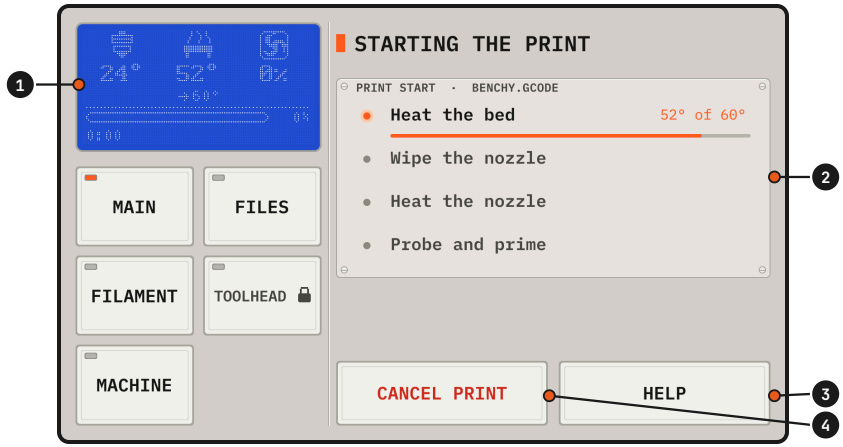
1. **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.
2. **– and +.** – moves the nozzle closer to the bed, + away from it. A line that is too round and thin needs –. A line that is flat and rough needs +.
3. **Z offset.** The height of the nozzle above its normal height. Watch the first lines while they print.
4. **Save.** Writes the Z offset into the config of the printer. Save config in Maintenance then writes the file.
5. **Step.** How far one tap moves the nozzle, in mm.
6. **Pause, Cancel.** Pauses the print, or stops it after a question.
7. **Help.** Opens Wi-Fi and help: the network, the address and the password of the deck.
8. **Layer.** The first layer of the model.

Watch the print

During a print, the deck shows the screen of the phase of the print. The deck knows the phase from the printer and from the G-code of the file.

1. Watch the first layer. Use − and + to set the height of the nozzle while it prints.
2. Tap Save to keep the Z offset for the next prints.
3. Use Tune to change the speed, the flow or the fan while the print runs.
4. When the print is done, wait for the bed to cool, then take the part off.

Preparing



1. **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.
2. **Steps.** The steps of the start macro of your printer, in its order. The deck follows the commands that the printer runs: the step that runs has a lit LED, and a done step has a dot.
3. **Help.** Opens Wi-Fi and help: the network, the address and the password of the deck.
4. **Cancel print.** Stops the print, after a question.

Send a file from the slicer

The deck answers the OctoPrint API that slicers use to send files and to start prints.

In Cura, add the deck with the OctoPrint Connection plugin. The deck asks "Allow Cura to use the Printdeck?". Tap Allow.

In OrcaSlicer or PrusaSlicer, add a printer of the type OctoPrint with the address http://printdeck.local, and the API key of the system page.

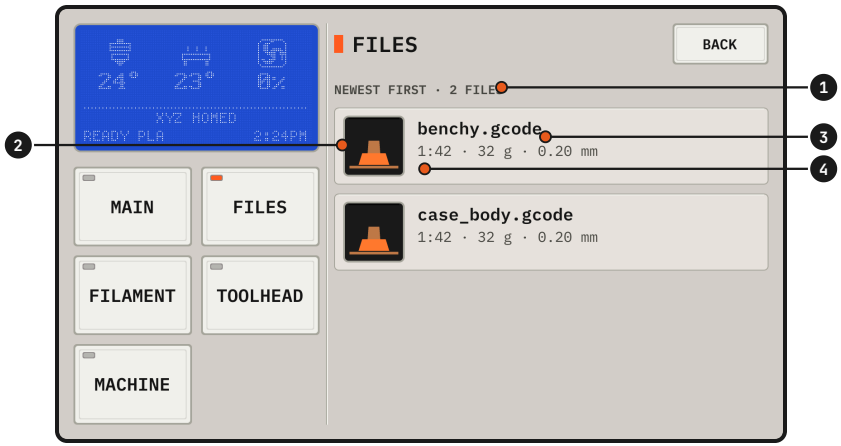
Start a print

The files are the G-code files that the slicer sent, or that you uploaded in the browser. The newest file is at the top.

Each file shows what the slicer wrote into it: the print time, the filament and the layer height. Cura, PrusaSlicer, OrcaSlicer and Bambu Studio write this data. A thumbnail shows when the slicer adds one.

1. Tap Print a file on the idle screen.
2. Tap the file. Check the print time, the filament and the temperatures.
3. Tap Print, then Print again.

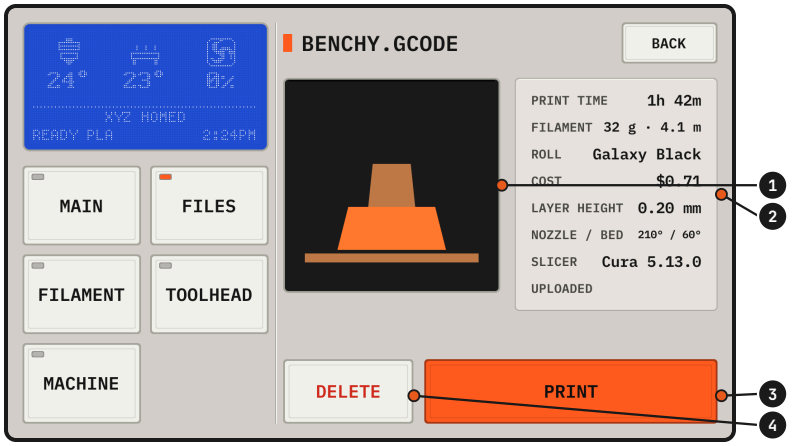
The file list



1. **Count.** The number of files.
2. **Picture.** The thumbnail of the slicer, or the shape of a part.
3. **File.** The name, the print time, the filament and the layer height.
4. **Date.** When the file arrived.

Drag the list up and down. Tap a file to open it.

A file



1. **Picture.** The thumbnail of the file.
2. **Slicer data.** The print time, the filament, the layer height, the temperatures, the slicer and the date.
3. **Print.** Starts the print, after a question.
4. **Delete.** Removes the file, after a question. The file that prints now cannot be removed.